

INPUT WINDOW

48 h × 26 feat.: lags, met,
reanalysis, AOD, calendar,
masks + station id



ENCODER

LSTM + station emb.
→ latent z_t



QUANTILE HEAD

monotone,
non-crossing
 $Q_\tau(H_{t+h})$



PREDICTIVE DIST.

CDF/PDF of
 $H_{t+1..24}$;
 $P(H \geq \text{thr})$

★ deliverable

PHYSICS HEAD

NO-NO₂-O₃
box-ODE; Leighton
+ O_x cons.



COMPOSITE LOSS

pinball + $\lambda_{\text{phys}} r$
+ $\lambda_{\text{chem}}(r_L + r_{O_x})$

regulariser, not accuracy